Experiment - 2

Aim :- Convert the ER diagram into relational schema. Write MYSQL CREATE TABLE statements with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL constraints. Insert sample data demonstrate referential integrity violations.

Source Code :

```
DROP DATABASE IF EXISTS ecommerce_db;
CREATE DATABASE ecommerce_db;
USE ecommerce_db;
```

⇒ 1. CUSTOMER [Strong entity]

```
CREATE TABLE Customers (
    customer_id  INT AUTO_INCREMENT
                 PRIMARY KEY,
    first_name   VARCHAR(50) NOT NULL,
    last_name    VARCHAR(50) NOT NULL,
    email        VARCHAR(50) NOT NULL UNIQUE
) ENGINE = InnoDB;
```

-- Multi-valued attribute : Customer Phone numbers

```
CREATE TABLE Customer_Phone (
    customer_ID INT NOT NULL
    phone VARCHAR(15) NOT NULL,
```

```
PRIMARY KEY (customer_ID, phone),
FOREIGN KEY (customer_ID) REFERENCES
                Customer (customer_ID)
        ON DELETE CASCADE
) ENGINE = InnoDB;
```

=> 2. ADDRESS (Weak entity - depends on Customer)

```
CREATE TABLE Address (
        customer_id INT NOT NULL
        address_id INT NOT NULL, --partial key
        street VARCHAR(100) NOT NULL,
        city VARCAR(50) NOT NULL,
        state VARCHAR(50) NOT NULL,
        pincode INT NOT_NULL,
PRIMARY KEY (customer_id, address_id), -- Composite key of weak attribute
FOREIGN KEY (customer_id) REFERENCES
        Customer (Customer_id) ON DELETE CASCADE
) ENGINE = InnoDB;
```

=> 3. CATEGORY (Strong Entity)

```
CREATE TABLE Category (
        category_id INT AUTO_INCREMENT PRIMARY KEY,
        category_name VARCHAR(50) NOT NULL UNIQUE
) ENGINE = InnoDB;
```

⇒ 4. SELLER (Strong entity)

```
CREATE TABLE Seller (
        seller_id INT AUTO_INCREMENT PRIMARY KEY,
        seller_name VARCHAR(50) NOT NULL,
    email VARCHAR(100) NOT NULL UNIQUE

) ENGINE = InnoDB;

CREATE TABLE Seller_Phone (
        seller_id INT NOT NULL,
    phone VARCHAR(15) NOT NULL,
    PRIMARY KEY (seller_id, phone),
    FOREIGN KEY (seller_id) REFERENCES
                seller (seller_id)
            ON DELETE CASCADE

) ENGINE = InnoDB;
```

⇒ 5. PRODUCT (Strong entity, with specialization)

```
CREATE TABLE Product (
    product_id INT AUTO_INCREMENT PRIMARY KEY,
    product_name VARCHAR(150) NOT NULL,
    price DECIMAL(10,2) NOT NULL,
    Stock INT NOT NULL DEFAULT 0,
    category_id INT NULL,
    Seller_id INT NOT NULL,
FOREIGN KEY (category_id) REFERENCES Category(category_id)
            ON DELETE SET NULL,
```

```
    FOREIGN KEY (Seller_id) REFRENCES Seller(Seller_id)
        ON DELETE CASCADE

) ENGINE = InnoDB ;

-- Specialization : Electronics (disjoint Subtype of Product)
CREATE TABLE Electronics (
        product_id  INT PRIMARY KEY,
        warranty_period  INT ,        -- in months
        brand VARCHAR (50)
    FOREIGN KEY (product_id) REFRENCES
            Product (product_id)
        ON DELETE CASCADE

) ENGINE = InnoDB;

-- Specialization : Clothing (disjoint Subtype of Product)
CREATE TABLE Clothing (
        product_id INT PRIMARY KEY,
        Size  VARCHAR *(10),
        fabric VARCHAR (50),
        FOREIGN KEY (product_id) REFRENCES
            Product (product_id)
        ON DELETE CASCADE

) ENGINE = InnoDB;
```

=> 6. ORDER (Strong Entity) - "Order" is a reserved word in MySQL

```
CREATE TABLE Orders (
        Order_Id INT AUTO_INCREMENT
                                PRIMARY KEY,
        order_date DATE NOT NULL,
        total_amount DECIMAL(10,2) NOT NULL,
        customer_id INT NOT NULL,
FOREIGN KEY (Customer_id) REFRENCES Customer
        (Customer_id) ON DELETE CASCADE
) ENGINE = InnoDB;
```

=> 7. Order_ITEM (Weak Entity - depends on orders)

```
CREATE TABLE Order_Item (
        order_id INT NOT NULL,
        Item_no INT NOT NULL, --partial key
        quantity INT NOT NULL,
        unit_price DECIMAL(10,2) NOT NULL,
        product_id INT NULL,
PRIMARY KEY (order_id, Item_no.), --composite
                                key of weak entity
FOREIGN KEY (product_id) REFRENCES Product
        (product_id) ON DELETE SET NULL
) ENGINE = InnoDB;
```

⇒ 8. PAYMENT (Strong entity, 1:1 with orders)

```
CREATE TABLE Payment (
    payment_id INT AUTO_INCREMENT PRIMARY KEY,
    amount DECIMAL(10,2) NOT NULL,
    payment_mode VARCHAR(20) NOT NULL,
    payment_date DATE NOT NULL,
    order_id INT NOT NULL UNIQUE,
    -- UNIQUE enforces the 1:1 cardinality
    FOREIGN KEY (order_id) REFERENCES orders(order_id)
        ON DELETE CASCADE
) ENGINE = InnoDB;
```

⇒ 9. DELIEVERY (Strong entity, 1:1 with orders)

```
CREATE TABLE Delievery (
    delievery_id INT AUTO_INCREMENT PRIMARY KEY,
    delivery_date DATE NOT NULL,
    Status VARCHAR(20) NOT NULL,
    tracking_number VARCHAR(30) UNIQUE,
    order_id INT. NOT NULL UNIQUE,
    FOREIGN KEY (Order_Id) REFERENCES (orderId)
        ON DELETE CASCADE
) ENGINE = InnoDB;
```

-- Sample Data

```
INSERT INTO Customer (First_name, last_name, email) VALUES
('Aarav', 'Sharma', 'aarav1230@example.com'),
('Priya', 'Singh', 'priya321@example.com');

INSERT INTO Customer_Phone (Customer_id, Phone) VALUES
(1, '8715705911')
(2, '9131578112')
(2, '8840495621')

INSERT INTO Address (customer_id, address_id,
street, city, state, pincode) VALUES
(1, 1, '43 BM Road', 'Lucknow', 'Uttar Pradesh',
226001),
(2, 1, '31 RC Road', 'Mumbai', 'Maharashtra',
'201161');

INSERT INTO Category (Category_name) VALUES
('Electronics'), ('Clothing'), ('Groceries');

INSERT INTO Seller (Seller_name, email) VALUES
('TechBazar Pvt Ltd', 'contact@techbazar.com'),
('Fashionhub India', 'support@fashionhub.com');
```

```
INSERT INTO Seller_Phone (Seller_id, phone) VALUES
(1, '011-40001234'),
(2, '022-50005678');

INSERT INTO Product (product_name, price, stock, category_id,
seller_id) VALUES
('Wireless Earbuds X200', 2499.00, 50, 1, 1),
('Cotton Kurta - Blue', 899.00, 100, 2, 2);

INSERT INTO Clothing (product_id, size, fabric) VALUES
(2, 'M', 'Cotton');

INSERT INTO orders (order_date, total_amount,
customer_id) VALUES
('2026-08-02', 3391.00, 1);

INSERT INTO Electronics (Product_Id, warranty_period, brand)
VALUES
(1, 12, 'Soundwave');

INSERT INTO Order_Item (order_id, item_no, quantity,
unit_price, product_id) VALUES
(1, 1, 1, 3000.00, 1),
(1, 2, 1, 899.00, 2);

INSERT INTO Payment (amount, payment_date, payment_mode,
order_id) VALUES
(4599.00, '2026-08-02', 'UPI', 1);
```

INSERT INTO Delivery (delivery_date, status, tracking_number, order_id) VALUES
('2026-08-06', 'Shipped', 'TRK123456', 1);

-- Demonstrating Referential Integrity
-- (a) INSERT violation: order_id 999 does not exist in Orders -- Expected: Error 1452 - cannot add or update a child row: a foreign key constraint fails (order_id, FK)
-- INSERT INTO Order_Item (order_id, item_no., quantity, unit_price, product_id)
-- VALUES (999, 1, 2, 500.00, 1);

-- (b) INSERT violation: product_id 999 does not exist in Product
-- Expected: Error 1452 - foreign key constraint fails (product_id FK)
-- INSERT INTO Order_Item (order_id, item_no, quantity, unit_price, product_id)
-- VALUES (1, 3, 1, 10000, 999);

-- (C) UNIQUE violation: This order already has a payment (1:1 constraint)
-- Expected: Error 1062 - Duplicate entry '1' for key 'order_id'
-- INSERT INTO Payment (amount, payment_date, payment_mode, order_Id)

--VALUES (500.00, '2026-08-07', 'cash', 1);

--(d) ON DELETE SET NULL demonstration:
delete a category.
-- Expected: Product.category_id for its products
becomes NULL, 1 row affected
-- DELETE FROM category WHERE category_id = 1;
-- SELECT * FROM Product WHERE product_id = 1;
-- category_id is now NULL;

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--(e) ON DELETE CASCADE demonstration: delete a customer
-- Expected: This Address, Customer_Phone, Orders rows are deleted automatically, which in turn cascades to order_item, Payment, and Delivery for those orders.
-- DELETE FROM Customer WHERE customer_id = 1;
-- SELECT * FROM Orders WHERE customer_id = 1;
-- empty result
-- SELECT * FROM Payment WHERE order_id = 1;
-- empty result

(f) NOT NULL violation: order_date is required
-- Expected: Error 1364 - Field 'order_date' doesn't have a default value.
-- INSERT INTO order (total_amount, customer_id)
VALUES (500.00, 1);